May 8, 1963

SUPERSEDING
GGG-B-00376a(GSA-FSS)
March 20, 1961 and
Fed. Spec. GGG-B-376
November 5, 1936

FEDERAL SPECIFICATION

BIT, SCREWDRIVER, (TAPERED SQUARE SHANK. FLAT TIP AND CROSS TIP)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers flat tip and cross tip screw-driver bits for driving plain slot and cross slot screws respectively. The bits are used in ratchet braces covered by Federal Specification GGG-B-671.

1.1.1 *Federal specification coverage.* This specification covers only the types, classes, and sizes of the commodity as generally purchased by the Federal Government, and does not include all types, etc., which are commercially available.

1.2 Classification.

1.2.1 *Types and classes.* Screwdriver bits shall be furnished in the following types and classes, when applicable, as specified (see 6.1):

Type I.—Flat tip (see figure 1).

Type II.—Cross tip.

Class 1.—Phillips type (figure 2).

Class 2.—Reed and Prince type (figure 3).

2. APPLICABLE SPECIFICATIONS, AND STANDARDS

2.1 Specifications and standards. The following specifications and standards of the issue in effect on date of invitations for bid form a part of this specification:

Federal Specifications:

GGG-B-671—Brace, Bit, Ratchet; Brace, Bit, Corner and Extensions, Bit.

GGG-S-121—Screwdriver and Screw Starter, Hand.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

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books and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-H-15424—Hand Tools; Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

MIL-STD-130—Identification Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The screwdriver bits shall be made of high carbon or alloy steel of a quality that will meet the test requirements herein. The screwdriver bit shall be free from brittleness or other defects likely to impair its serviceability.

3.2 Method of manufacture. The bit shall be manufactured in a manner to meet the test requirements herein, and shall be free from defects or flaws which may adversely affect its serviceability or appearance.

3.3 Shank. The bit shall have a tapered square shank suitable for use in chucks on ratchet braces in accordance with GGG-B-671.

3.4 Hardness. The Rockwell C hardness of the blade end of the bit shall be not less than 50 nor more than 60. The hardness test on flat tip bits shall be taken within 1/4 inch of the extreme end of the blade, and on cross-tip bits shall be taken on the round portion of the shank as near to the point end as possible to receive an accurate reading.

3.5 Straightness. The screwdriver bit shall be straight so that it will run concentrically when properly chucked and operated in a ratchet brace.

3.6 Finish. The bit shall be accurately finished to size to provide uniformity and proper fit in the screw slots. The convex portions of type I bits, and the cylindrical portion of all bit shanks shall be full-polished or buffed, and these surfaces after final finishing operation(s) shall have a 30-microinch finish or better.

3.7 Marking. Each bit shall be marked in a plain and permanent manner with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined. For military purchases each bit shall be marked for identification in accordance with MIL-STD-130.

3.8 Type I. Type I screwdriver bits (figure 1) shall have flat tips and shall be in accordance with the requirements of table I.

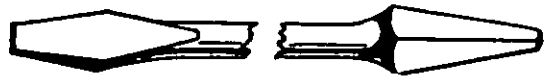


Figure 1. - Type I. - Flat tip screwdriver bit.

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Table I. - Requirements for type I flat tip screwdriver bits.

Size nominal width of blade at end ± .031	Thickness of blade at end (nominal) ± .005	Overall length of bit ± .500	Diameter of body (minimum)	Torsional strength (minimum)
Inch	Inch	Inches	Inch	Inch-pounds
3/16	0.030	5	3/16	25
1/4	.037	5	3/16	50
5/16	.041	5	1/4	100
3/8	.047	5	5/16	175
1/2	.064	5	5/16	350
5/8	.070	5	5/16	500
3/4	.080	5	3/8	650

3.8.1 *Blade.* All four sides and the end of flat tip bits shall be ground to size with 120-grit or finer abrasive, resulting in a 40-micro-inch finish or better, and shall be symetrical.

3.9 Type II. Type II screwdriver bits shall have cross tips.

3.9.1 *Class* 1. Type II, class 1, cross tip screwdriver bits shall have a Phillips type point (figure 2) and shall be in accordance with the requirements of table II. The point end of the bits shall be in accordance with the dimensions shown for the respective point size numbers of Phillips type screwdrivers in GGG-S-121. The screwdriver bits shall be marked with the name "Phillips" and the point(s) size number.



Figure 2. - Type II, class 1, cross tip, Phillips type, screwdriver bit (For point details see Federal Specification GGG-S-121).

Table II. - Requirements for type II, class 1, cross tip, Phillips type screwdriver bits.

Point size	Overall length of bit ± .500	Torsional strength (minimum)
Number	Inches	Inch-pounds
1	4-1/2	35
2	4-1/2	70
3	4-1/2	150
4	4-1/2	200

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Figure 3. - Type II, class 2, cross tip, Reed and Prince type screwdriver bit. (For point details see Federal Specification GGG-S-121.

Table III. - Requirements for type II, class 2, cross tip, Reed and Prince type screwdriver bit.

Diameter ± .010	Overall length of bit ± .500	Torsional strength minimum
Inch	Inches	Inch-pounds
5/16	4-5/8	150

3.9.2 *Class* 2. Type II, class 2, cross tip screwdriver bit shall have a Reed and Prince type point (figure 3) and shall be in accordance with the requirements of table III. The point end of the bits shall be in accordance with the dimensions shown for the respective diameters of Reed and Prince screwdrivers in GGG-S-121. The screwdriver bits shall be marked with the name "Reed and Prince" or the abbreviation "R & P". (Type II, class 2 screwdriver bits manufactured by "Reed and Prince" and marked with their trademark will meet this marking requirement.)

3.10 **Workmanship.** Standards of workmanship shall be such that bits specified herein shall comply with the requirements of sections 3 and 4. Workmanship shall be of such quality as to assure a product free of defects such as bulge, warpage, burrs, rust, undesirable sharp edges, cracks, scale, or damaged bits.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Inspection responsibility.** The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Sampling for inspection and acceptance.** Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

4.2.1 *Inspection of materials and components.* In accordance with 4.1, the supplier is responsible for insuring that components and materials used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2.2 *Inspection of the end item.* Inspection of the bits shall be performed as specified in the following paragraphs. A lot shall

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consist of bits of the same type, class and size. The sample unit shall be one completely fabricated and assembled bit.

4.2.2.1 *Visual examination.* Each sample unit shall be visually examined for the defects listed in table IV. The inspection level shall be level II with an acceptable quality level of 4.0 defects per 100 units.

Table IV.—Classification of defects.

Examination	Defects
Finish	Indication of rust. Not accurately finished to provide proper fit in screw slot.
Design	Not type and class specified.
Material	Not high carbon or alloy steel.
Workmanship	Not free from pits, nodules, cracks, dents, or other detrimental defects. Shank not tapered. Bit not straight. Tips not symmetrical. Corners and edges not rounded. Grind or chatter marks, deep scratches or other indications of poor workmanship.
Marking	Missing, incorrect, not legible.

4.2.2.2 *Dimensional examination.* Each sample unit shall be examined for dimensions specified in applicable paragraphs and tables. Any nonconformance with requirements shall constitute a defect. The inspection level shall be level L6, with an acceptable quality level of 4.0 defects per 100 units.

4.2.2.3 *Testing.* Each sample unit shall be tested as specified in 4.3 through 4.4. Any determination not within specified requirements shall constitute a defect. The inspection level shall be level L4, with an acceptable quality level of 1.5 defects per 100 units.

4.2.3 *Inspection of preparation for delivery.* Unless otherwise specified, inspection to determine compliance with preparation for delivery requirements shall be accomplished in accordance with MIL-P-15424.

4.3 Test procedures.

4.3.1 *Torque test.* One end of the bit shall be held securely while a torsional strain is applied. The torsion is obtained by multiplying the force in pounds by the perpendicular distance, in inches, between the point where the force is applied to the lever and the longitudinal axis of the bit. This value shall be expressed in inch-pounds. Approved equipment for making this test is shown in illustrations, figures 4 and 5. The equipment illustrated in figure 4 is suitable where a lathe is available. The equipment illustrated in figure 5 is a small portable device suitable for clamping to a work bench. Other suitable equipment, as maintained by the manufacturers, capable of producing equivalent results, may be used. The blade of the type I flat-tip-screwdriver bits shall be held in plugs such as illustrated in figures 4 and 5. These plugs shall have a slot to receive the blade. The depth of this slot shall not exceed one and one-half times the thickness of the end of the blade. The width of the slot shall exceed the thickness of the end of the blade by not more than 1/64 inch. The blade of type II cross slot bits shall be held in plugs such as illustrated in figures 4 and 5 except that in lieu of a plain slot, a cross slot conforming to the "Phillips" type or "Reed and Prince" type as applicable, and as defined in GGG-S-121 shall be employed. The chuck may be taken from an ordinary bit brace, the bow being cut off and an extension welded into place. A weight shall be hung on the lever and the location of the weight for the desired torsion should be marked on the lever. The lever shall be horizontal after the weight is applied. When the torsion is applied, the screwdriver bit shall not break and after the torsion is removed the bit shall show no measurable permanent deformation. By employing a ratchet brace to apply the torque, the torque test equipment shown in GGG-S-121 may be employed in lieu of the test equipment shown in figures 4 and 5.

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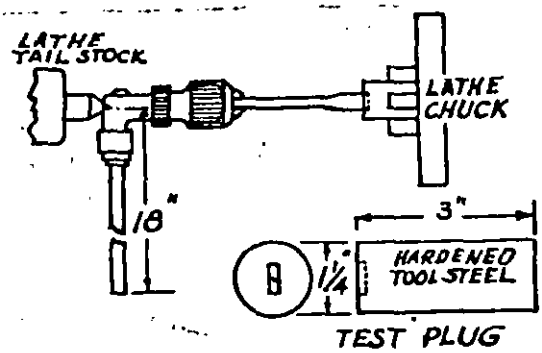


Figure 4. - Torque testing equipment used in a lathe.

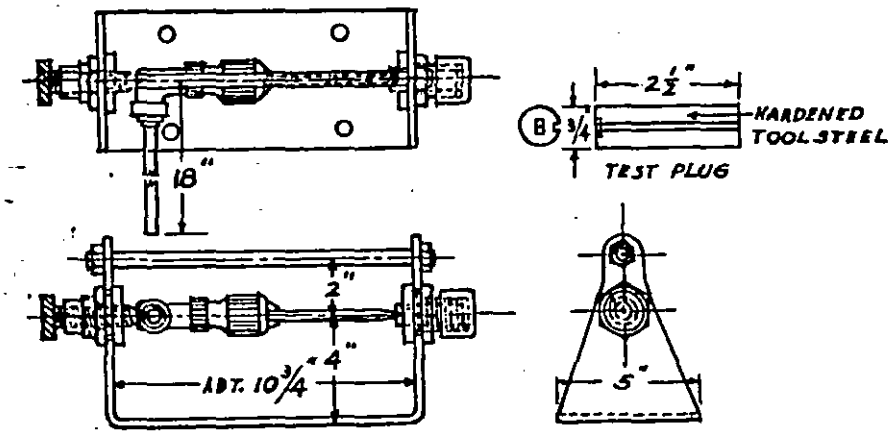


Figure 5. - Torque testing equipment used on a bench.

4.4 Hardness test. Bits shall be hardness tested to determine conformance with 3.4. Failure of any bit to meet the hardness test shall be cause for rejection of the lot represented by the sample.

5. PREPARATION FOR DELIVERY

(Civil agencies. Federal Standard No. 102 should be referred to for definitions and application of the various levels of packaging protection for supplies and equipment.)

5.1 Cleaning, preservation, and packaging.

5.1.1 *Level A.* Level A cleaning, preservation, and packaging shall be in accordance with level A in MIL-H-15424.

5.1.2 *Level C.* Level C cleaning, preservation, and packaging shall be in accordance with the commercial practice.

5.2 Packing.

5.2.1 *Level A.* Level A packing shall be in accordance with level A in MIL-H-15424.

5.2.2 *Level B.* Level B packing shall be in accordance with level B in MIL-H-15424.

5.2.3 *Level C.* Screwdriver bits preserved and packaged in accordance with 5.1.1 and 5.1.2 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with the rules or regulations of carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any spe-

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cial marking required by the contract or order (see 6.1), shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military.* In addition to any special marking required by the contract or order (see 6.1), shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type, class, and size required (1.2.1 and 1.2.2 and tables I, II, and III).

(*c*) Levels of packaging and packing required (section 5).

(*d*) Marking, if different (5.3).

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Copies of this specification may be purchased for 5 cents each.

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